

EE645 — Physical Models in Remote Sensing

Final Report: Discrete Ordinates Method

Vegetation Canopy Radiative Transfer
DOM_v3 — Bi-Lambertian Phase Function

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Energy Balance Table

Energy Balance Table — DOM_v3
fdir=1.0 (pure direct beam → DHR), SZA=40°, N=16, M=16, K=50, uniform LAD (G=0.5)

Configuration	DHR	Transmittance	Canopy Abs.	Ground Abs.	Imbalance (%)
RED LAI=1.5	0.0318	0.3887	0.6185	0.3498	-0.013%
RED LAI=4.0	0.0223	0.0793	0.9065	0.0713	-0.020%
NIR LAI=1.5	0.3214	0.6359	0.1330	0.5405	+0.512%
NIR LAI=4.0	0.4177	0.3100	0.3089	0.2635	+0.987%

DHR = Directional-Hemispherical Reflectance (= BHR when fdir=1.0). Transmittance = $F_{\downarrow_ground} / F_{in}$. Imbalance = $F_{in} - (DHR + Canopy\ Abs. + Ground\ Abs.)$.

Figure 1 — Leaf Angle Distributions

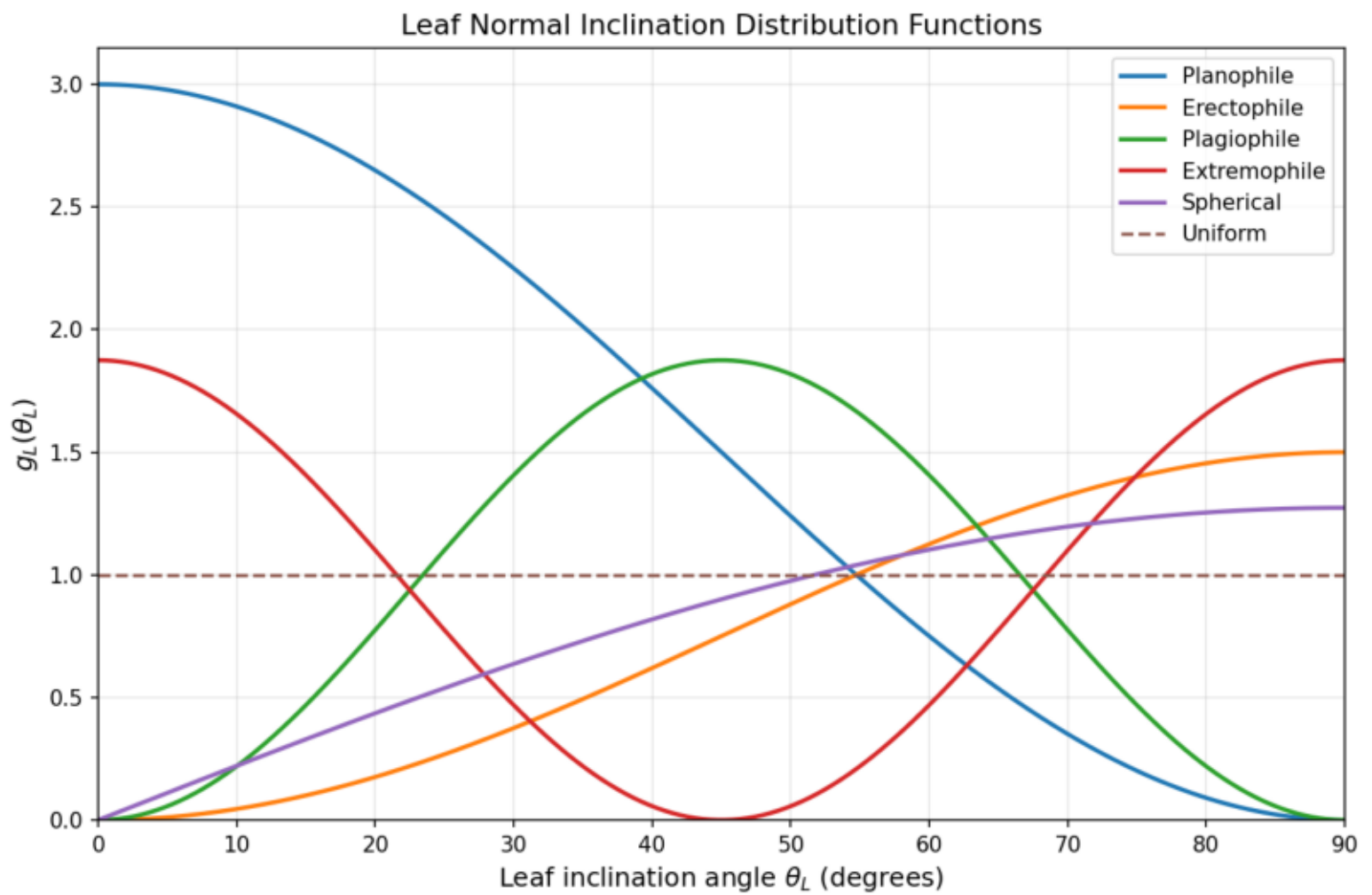


Figure 2 — G-Function for Various Leaf Angle Distributions

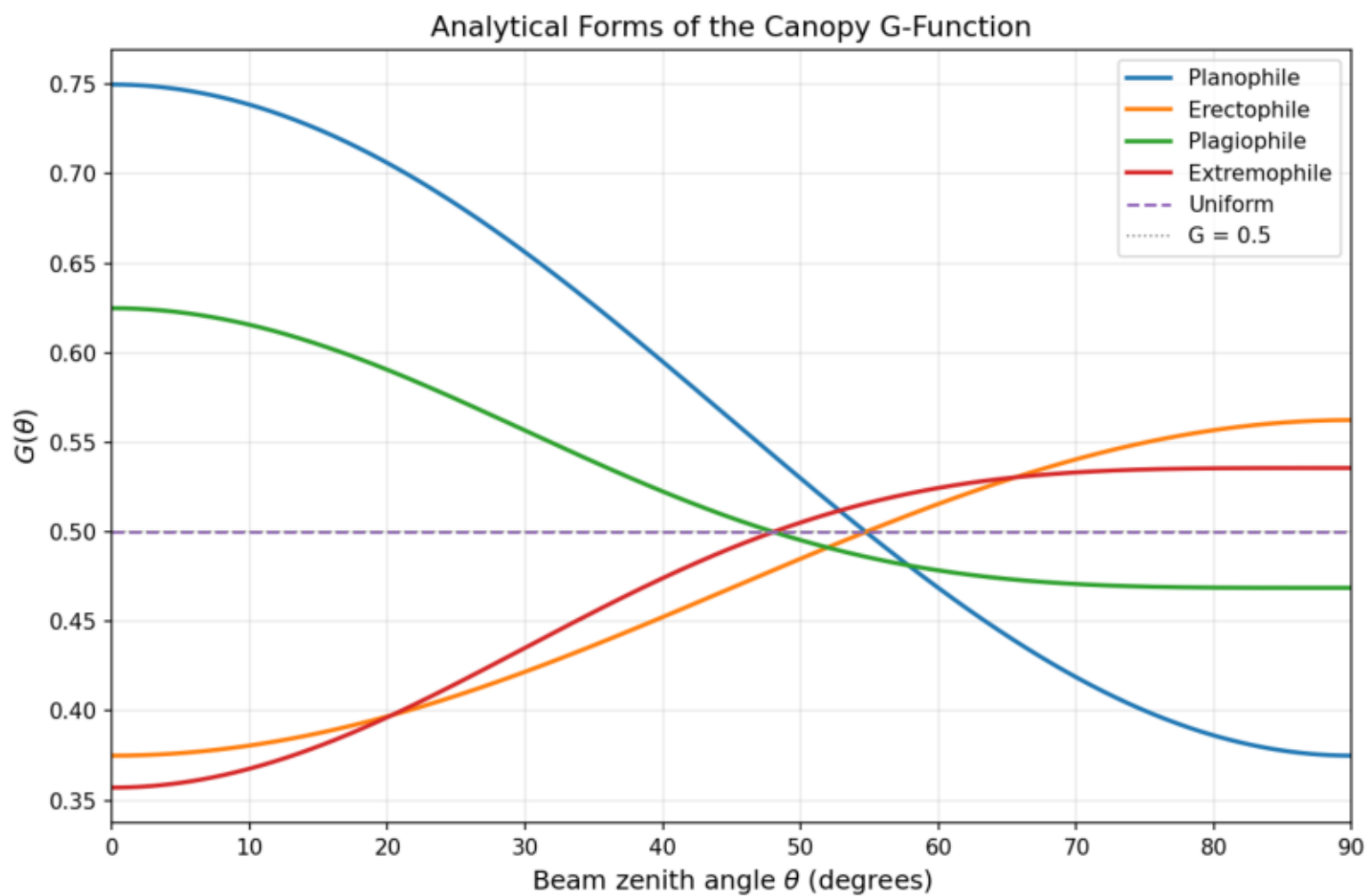


Figure 3 — Volume Scattering Phase Function

$\Gamma_{LD,U}$ for Uniform Leaf Inclination and Azimuthal Distribution
($\tilde{g}_L = 1$, $h_L = 1$, $\omega_{LD} = 1$)

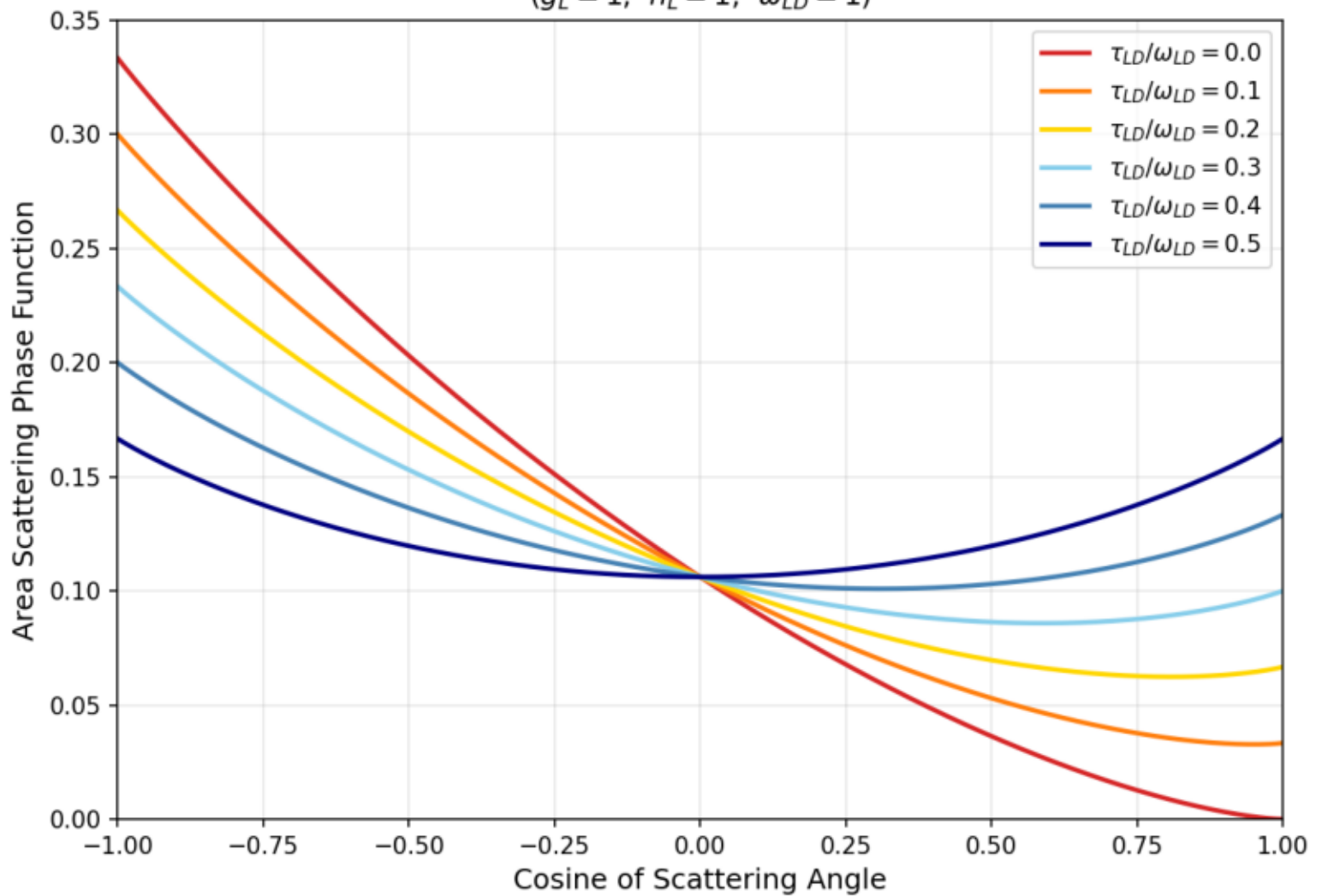


Figure 4 — Uncollided Radiation: RED Band

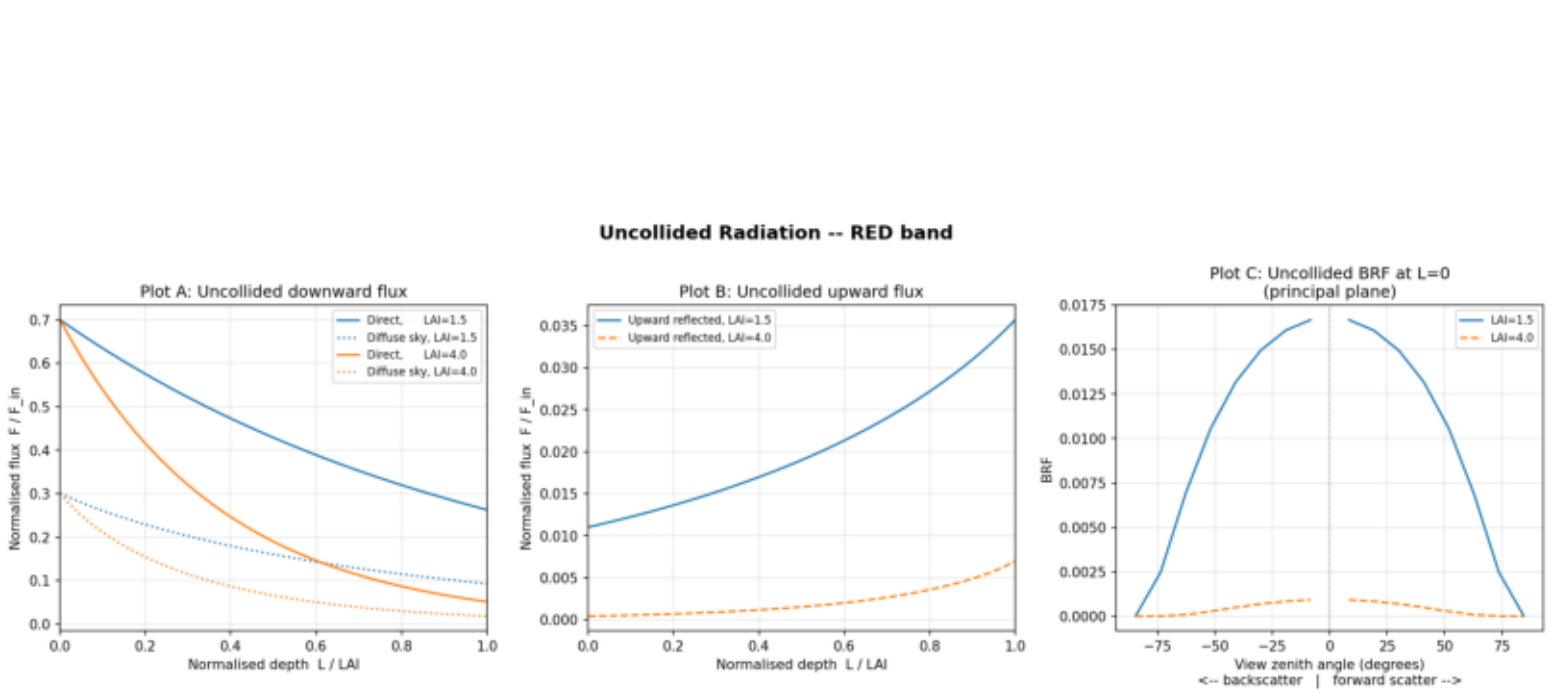


Figure 5 — Uncollided Radiation: NIR Band

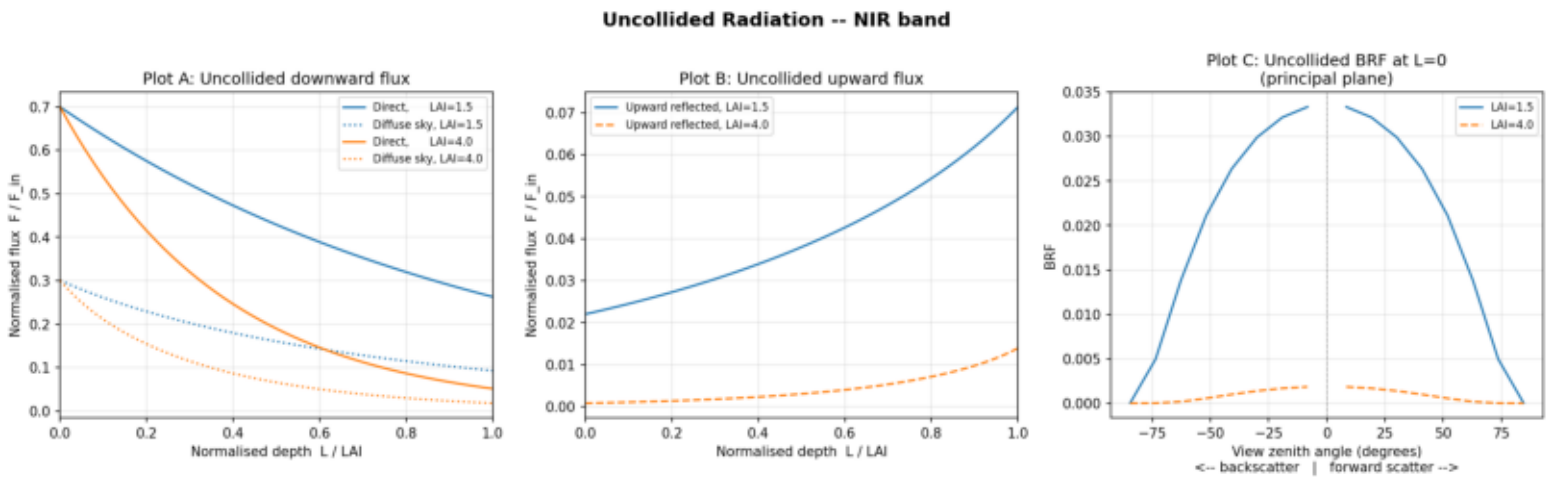


Figure 6 — HDRF Hemispherical Polar Map (fdir=0.70)

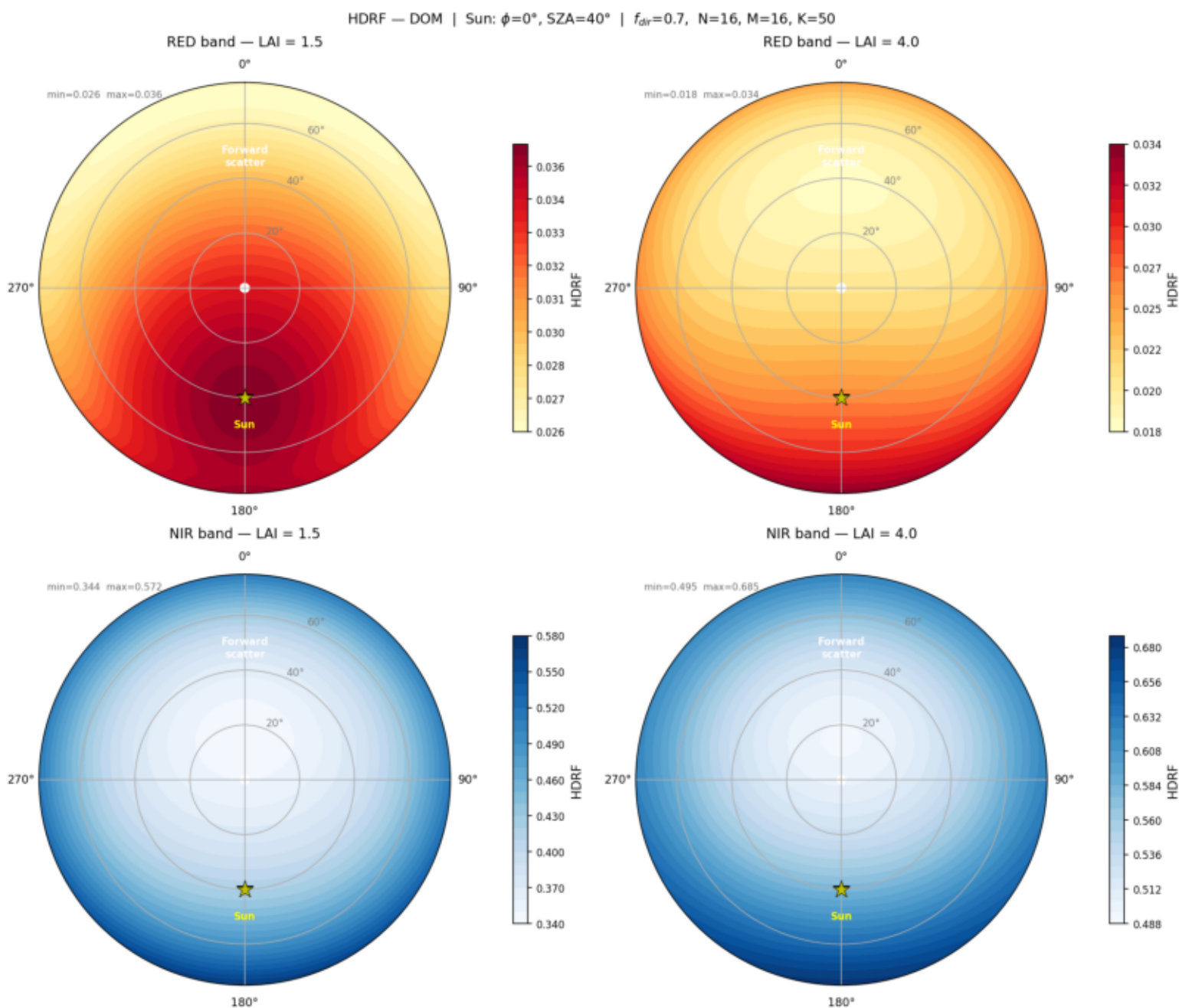


Figure 7 — HDRF Principal Plane Cross-Section (fdir=0.70)

HDRF — Principal Plane | $f_{dir}=0.7$, $SAZ=40^\circ$, $\phi_{solar}=0^\circ$, $N=16$, $M=16$, $K=50$

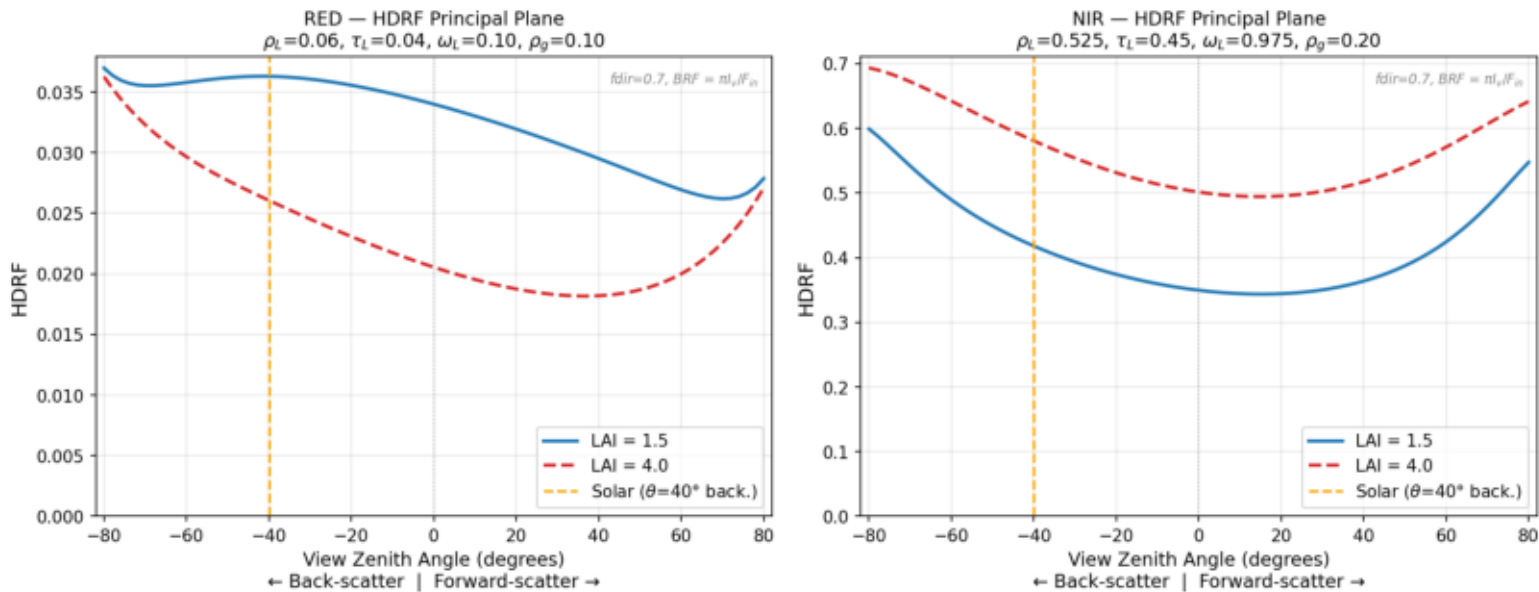


Figure 8 — BHR as a Function of LAI and Soil Brightness

Fig. 08 — BHR as a Function of LAI and Soil Brightness
SZA=40°, fdir=1.0, uniform LAD

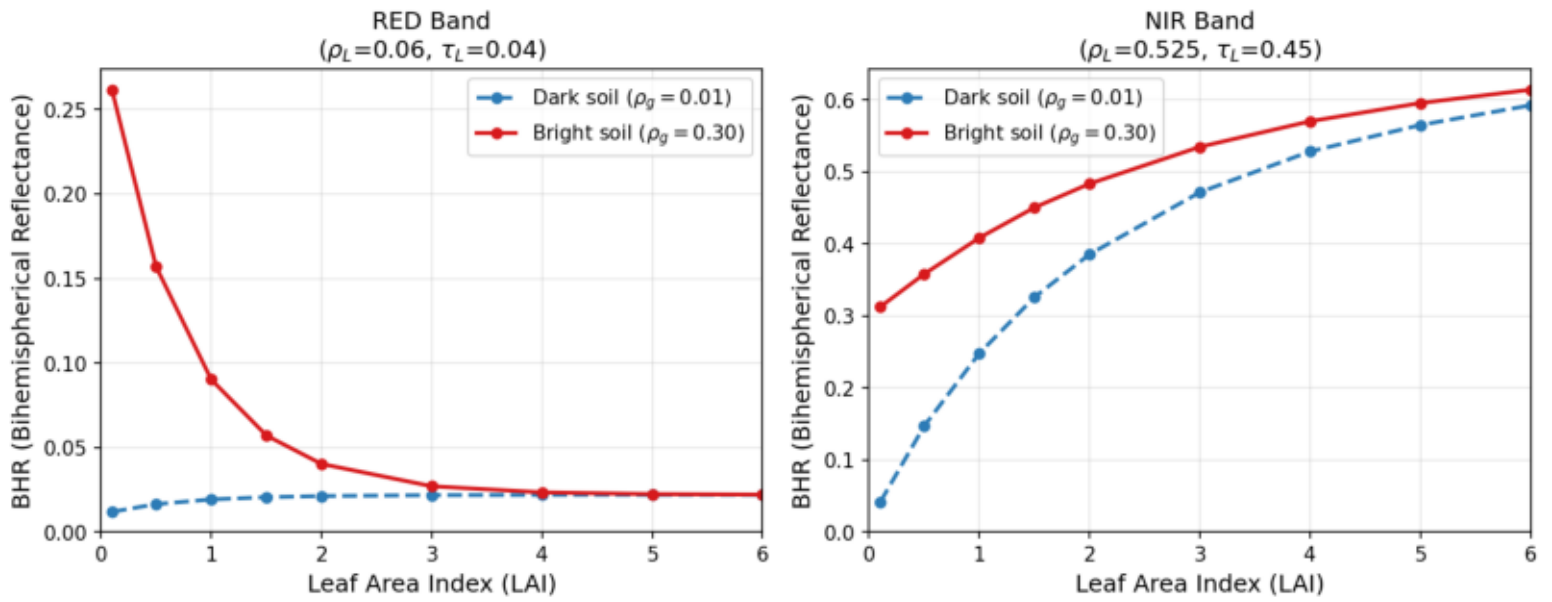


Figure 9 — LAI Effect on BHR and Transmittance

Fig. 09 — LAI Effect on BHR and Transmittance
SZA=40°, fdir=1.0, uniform LAD

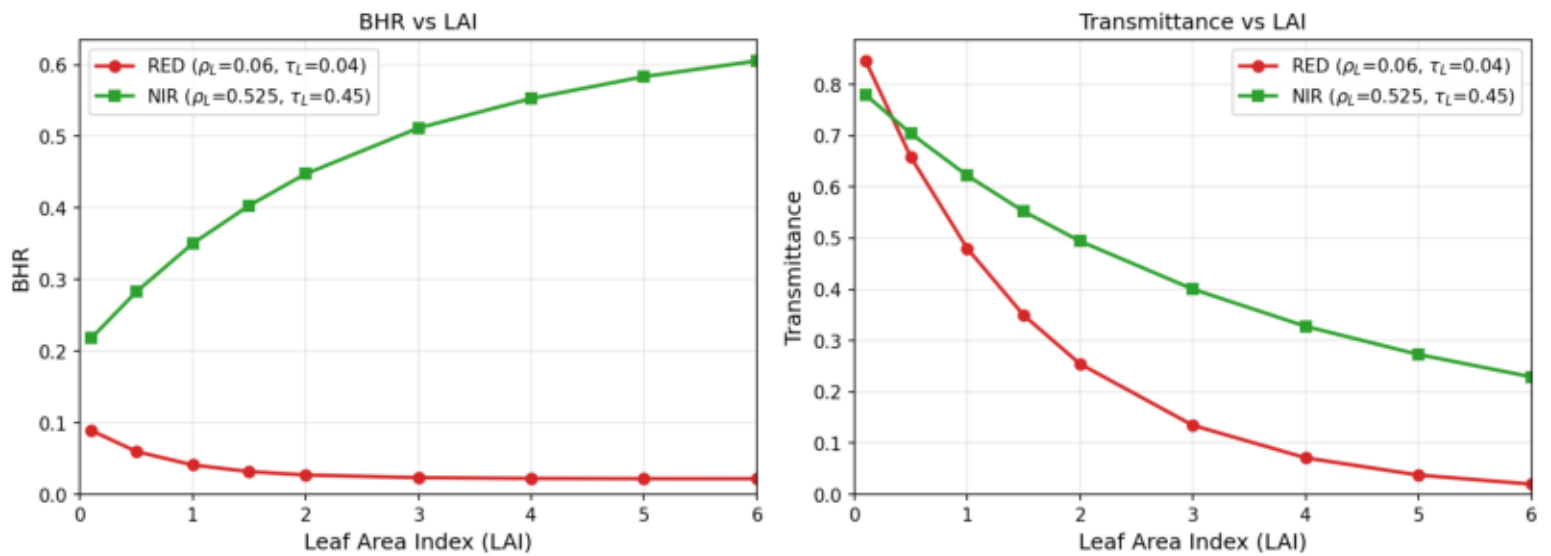


Figure 10 — Solar Polar Angle Effect on BHR and Transmittance

Fig. 10 — Solar Polar Angle Effect on NIR BHR and Transmittance
fdir=0.7, $\rho_g=0.2$, uniform LAD (G=0.5)

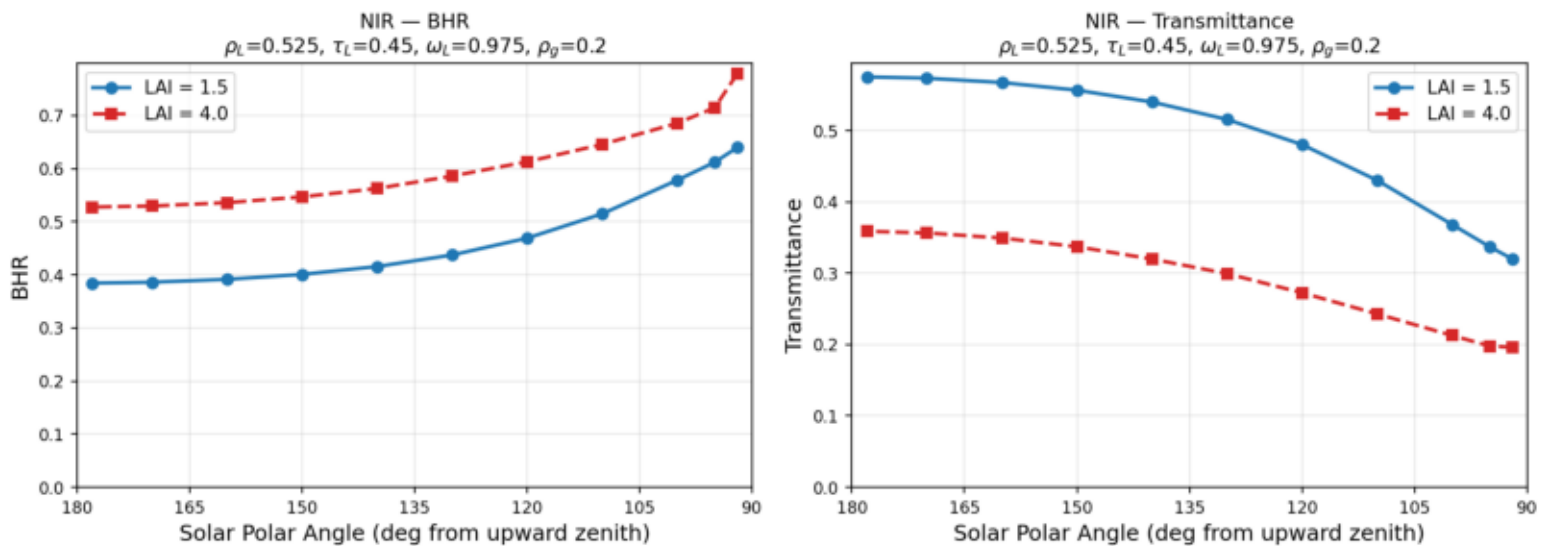
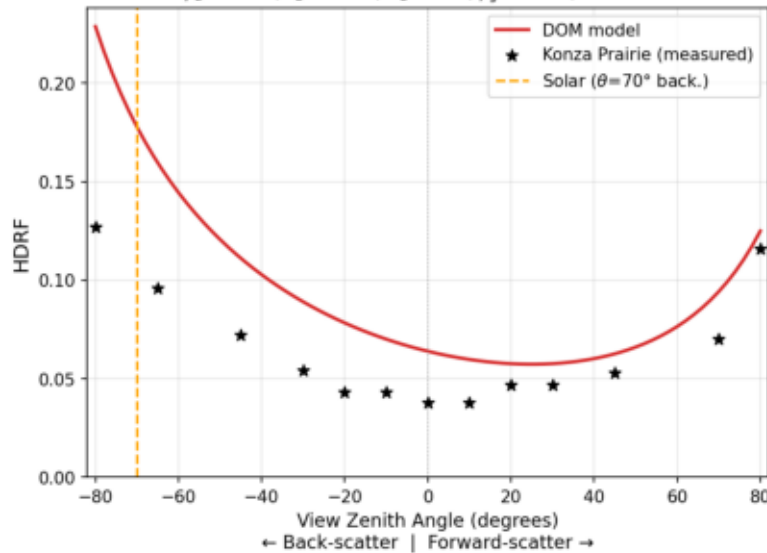


Figure 11 — Konza Prairie: DOM Model vs. Measured HDRF

HDRF — Principal Plane | Konza Prairie: DOM vs Measured | LAI=2.2, SZA=70°, N=16, M=16, K=50

RED — Principal Plane HDRF
 $\rho_L=0.1814, \tau_L=0.0926, \omega_L=0.274, \rho_g=0.0825, fdir=0.862$



NIR — Principal Plane HDRF
 $\rho_L=0.4525, \tau_L=0.4913, \omega_L=0.944, \rho_g=0.1363, fdir=0.931$

